



Enhancing the local power of IVX-based tests in predictive regressions



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HIGHLIGHTS

- In simple predictive models with persistent regressors, the extended instrumental variable method (IVX) stabilizes size, but loses power.
- Augmenting the test regression with the lagged dependent variable may seriously increase local power.
- The initial and the augmented IVX methods are equivalent under the null hypothesis of no predictability.
- The power gain of the augmented IVX method is maximal when the IVX method has smallest power.

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ABSTRACT

IVX estimation is used increasingly often in predictive regressions with regressors of unknown persistence. While not exhibiting the second-order bias the OLS estimator has in this setup, IVX estimators have reduced rates of convergence when the regressors are highly persistent. The reduced convergence rates may sometimes lead to power losses in finite samples when testing for no predictability, for instance. The note discusses a simple way of improving the local power of IVX-based tests, consisting of augmenting the predictive regression with the lagged dependent variable. This implies a feed-back loop which strengthens the signal of the IVX instrument without changing its dynamic properties. The proposed augmentation works best when the power loss of IVX would have been maximal compared to the infeasible OLS-based test.

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1. The inferential problem

Consider the popular predictive regression model

$$y_t = \beta x_{t-1} + u_t, \quad (1)$$

where the regressor has an autoregressive structure,

$$x_t = \rho x_{t-1} + v_t. \quad (2)$$

The typical predictor in such regressions has high, but unknown persistence, so it is assumed in line with the literature (e.g. Campbell and Yogo, 2006) that the autoregressive parameter is near to unity, $\rho = \rho_T = 1 - c/T$. To fix ideas, let us work without intercept in (1) for now, and the shocks u_t and v_t are assumed to be a zero-mean bivariate i.i.d. sequence with positive definite covariance matrix $(\sigma_u^2 \sigma_{uv}; \sigma_{uv} \sigma_v^2)$. This implies the weak convergence

$\frac{1}{\sqrt{T}} \sum_{t=1}^{\lfloor sT \rfloor} (u_t, v_t)' \Rightarrow (\sigma_u W(s), \sigma_v V(s))'$ with $(W(s), V(s))'$ a vector of correlated standard Wiener processes; hence x_t exhibits a stochastic trend, $\frac{1}{\sqrt{T}} x_{\lfloor sT \rfloor} \Rightarrow B_c(s)$, where B_c is the Ornstein–Uhlenbeck process driven by V , $B_c(s) = V(s) - c \int_0^s e^{-c(s-r)} V(r) dr$. To keep the regression balanced, the slope coefficient in (1) needs to be of small magnitude.

Should the shocks u_t and v_t be correlated, the regressor is said to be endogenous; given the high persistence of the regressor x_t , the OLS estimator of β is biased in finite samples (e.g. Stambaugh, 1999). This second-order bias affects the limiting distribution of t statistics for the null $\beta = 0$: Elliott and Stock (1994) show the t statistic t_β to possess a nonstandard limiting distribution depending on c whenever $\sigma_{uv} \neq 0$. Note however that c cannot be consistently estimated.

Among others, Campbell and Yogo (2006), Jansson and Moreira (2006), Rodrigues and Rubia (2011) or Breitung and Demetrescu (forthcoming) study persistence-robust inference in (1). IVX-based inference (Phillips and Magdalinos, 2007; Magdalinos and Phillips, 2009) has proved to be an attractive alternative. See e.g. Kostakis

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et al. (2010), Gonzalo and Pitarakis (2012), Lee (2012) or Phillips and Lee (2013). The IVX method uses $z_{t-1} = (1 - \phi L)^{-1} \Delta x_{t-1}$ as instrument for x_{t-1} , where $\phi = 1 - \eta/T^\eta$ with $\eta \in (0, 1)$ and $a > 0$ such that z_t is mildly, or moderately, integrated. This results in asymptotically chi-squared distributed Wald-type test statistics, irrespective of c . While the model here is that x_t is near integrated without short-run dynamics, note that the IVX procedure is more general and can be applied when x_t is mildly integrated, mildly explosive, or even stationary, and also under short-run dynamics.

For the IVX-based test, the improved size control under the null comes at the cost of reduced power under the alternative. It holds namely for the IVX estimator that $\hat{\beta}_{ivx} - \beta = O_p(T^{-\frac{\eta+1}{2}})$ (Magdalinos and Phillips, 2009); the limiting distribution of the IVX-based t type statistic t_{ivx} for the null $\beta = 0$ under sequences of local alternatives of the type $\beta = \beta_T = bT^{-\frac{\eta+1}{2}}$ is shown by Breitung and Demetrescu (forthcoming) to be

$$t_{ivx} \Rightarrow Z + b \frac{\sigma_v \sqrt{2}}{\sigma_u \sqrt{a}} \left(B_c^2(1) - \int_0^1 B_c(s) dB_c(s) \right)$$

with $dB_c(s)$ shorthand for $-cB_c(s)ds + dV(s)$ and Z a standard normal variate independent of $V(s)$ and $B_c(s)$.

In theory, η may be chosen arbitrarily close to 1 such that the power losses of the IVX-based test be reduced to a minimum. But, in finite samples, this comes at the cost of loosening size control, since a mildly integrated process with η close to 1 cannot reliably be distinguished from a near integrated one for small T . So the question addressed in this note is whether the power can be enhanced for small η .

2. Increasing the local power

Consider the following lag-augmented version of the predictive regression (1),

$$y_t = \gamma y_{t-1} + \beta x_{t-1} + u_t, \quad (3)$$

where $\gamma = 0$ but the restriction is not imposed when estimating. This is in a sense a variable addition procedure; cf. Dolado and Lütkepohl (1996) and Toda and Yamamoto (1995). The effect, however, is not to reduce the signal, as is the case with the original variable addition (see Breitung and Demetrescu, forthcoming, for a more recent discussion of the variable addition approach) but rather to amplify it, as can be seen in the following proposition which is proved in Section 3.

Proposition 1. Denote by $\hat{\beta}_{ivx}^*$ the IVX estimator of β in (3) – with y_{t-1} instrumented by itself – and by t_{ivx}^* the corresponding t type statistic, and assume i.i.d. shocks with finite fourth-order moments.

Let $\sigma_{uv} \neq 0$.

1. If $0 < \eta < 1/3$ then, for $\beta_T = bT^{-\frac{3-\eta}{4}}$,

$$t_{ivx}^* \xrightarrow{d} Z - b^2 \frac{\sqrt{2a}\sigma_{uv}\sigma_v}{\sigma_u^3} \int_0^1 B_c^2(s) ds.$$

2. If $\eta = 1/3$ then, for $\beta_T = bT^{-\frac{2}{3}}$,

$$t_{ivx}^* \xrightarrow{d} Z + \frac{b \frac{\sigma_v}{a} \left(B_c^2(1) - \int_0^1 B_c(s) dB_c(s) \right) - b^2 \sigma_{uv} \frac{1}{\sigma_u^2} \sigma_v^2 \int_0^1 B_c^2(s) ds}{\sigma_u \sqrt{\frac{\sigma_v^2}{2a}}}.$$

3. If $1/3 < \eta < 1$ then, for $\beta_T = bT^{-\frac{\eta+1}{2}}$,

$$t_{ivx}^* \xrightarrow{d} Z + b \frac{\sigma_v \sqrt{2}}{\sigma_u \sqrt{a}} \left(B_c^2(1) - \int_0^1 B_c(s) dB_c(s) \right).$$

Should $\sigma_{uv} = 0$, then, for $\beta = bT^{-\frac{\eta+1}{2}}$ and all $0 < \eta < 1$,

$$t_{ivx}^* = Z + b \frac{\sigma_v \sqrt{2}}{\sigma_u \sqrt{a}} \left(B_c^2(1) - \int_0^1 B_c(s) dB_c(s) \right).$$

Since $T^{-\frac{3-\eta}{4}}$ vanishes faster than $T^{-\frac{\eta+1}{2}}$ when $\eta < 1/3$, the inclusion of the lagged dependent variable has a positive effect on local power when η is closer to zero and $\sigma_{uv} \neq 0$; in other words, precisely when IVX is most needed and the power losses would have been large. At the same time, the limiting distribution is the same as that of the initial IVX-based test when $\eta > 1/3$ or $\sigma_{uv} = 0$ so nothing is lost if using the augmented IVX estimator in such situations. The results extend to the case with intercept in (1) along the lines of Kostakis et al. (2010).

For $\eta < 1/3$, the local power depends on b^2 rather than on b . The direction of the shift under local alternatives is given according to the proposition by the sign of the covariance σ_{uv} causing endogeneity. If focusing on two-sided tests, as is done in the following, this does not pose problems under the alternative. Note however that an adaptive procedure taking the (estimated) sign of σ_{uv} into account may further improve the power; this topic is not pursued here, though.

Finally, demeaning the series prior to IVX estimation of (1) and (3) does not affect the results qualitatively: the limiting distributions from Proposition 1 are given in terms of demeaned OU processes. In fact, as discussed by Kostakis et al. (2010, p. 14), it suffices to demean y_t and x_t . Not demeaning the instrument z_t is inconsequential in the limit and reduces the finite-sample distortions of the IVX t statistics; we thank the referee for pointing this out.

3. Proof

To prove the proposition for $\sigma_{uv} \neq 0$, note that, under the above assumptions,

$$\frac{1}{T^{1/2+\eta/2}} \sum_{t=2}^T z_{t-1} u_t \Rightarrow \mathcal{N} \left(0, \frac{\sigma_u^2 \sigma_v^2}{2a} \right) \quad (4)$$

$$\frac{1}{T^{1+\eta}} \sum_{t=2}^T z_{t-1}^2 \xrightarrow{p} \frac{\sigma_v^2}{2a} \quad (5)$$

$$\frac{1}{T^{1+\eta}} \sum_{t=2}^T z_{t-1} x_{t-1} \Rightarrow \frac{\sigma_v^2}{a} \left(B_c^2(1) - \int_0^1 B_c(s) dB_c(s) \right),$$

see Magdalinos and Phillips (2009) and Breitung and Demetrescu (forthcoming, Corollary 2). It is also known that, under the assumptions of the proposition,

$$\frac{1}{T} \sum_{t=2}^T x_{t-2} u_{t-1} = O_p(1),$$

$$\frac{1}{T} \sum_{t=2}^T x_{t-1} u_{t-1} = O_p(1) \quad \text{and} \quad \frac{1}{T^2} \sum_{t=2}^T x_{t-2}^2 = O_p(1).$$

One then obtains as $T \rightarrow \infty$ that

$$\begin{aligned} \frac{1}{T} \sum_{t=2}^T z_{t-1} y_{t-1} &= \frac{T^\eta \beta_T}{T^{1+\eta}} \sum_{t=2}^T z_{t-1} x_{t-2} \\ &\quad + \frac{1}{T} \sum_{t=2}^T (\phi_T z_{t-2} + v_{t-1}) u_{t-1} \xrightarrow{p} \sigma_{uv} \end{aligned}$$

since $T^\eta \beta_T = o(1)$ in all three cases and $\rho_T \leq 1$, and

$$\begin{aligned} \sum_{t=2}^T z_{t-1} y_t &= \beta_T \sum_{t=2}^T z_{t-1} x_{t-1} + \sum_{t=2}^T z_{t-1} u_t \\ &= O_p(\max \{ \beta_T T^{1+\eta}, T^{1/2+\eta/2} \}) = O_p(T^{1/2+\eta/2}). \end{aligned}$$

Table 1Rejection frequencies of no predictability tests under local alternatives $\beta_T = b/T$ when the regressor is integrated.

	b	$T = 100$									$T = 1000$								
		$\omega = -0.98$			$\omega = -0.5$			$\omega = 0$			$\omega = -0.98$			$\omega = -0.5$			$\omega = 0$		
		t_β	t_{ivx}	t_{ivx}^*	t_β	t_{ivx}	t_{ivx}^*	t_β	t_{ivx}	t_{ivx}^*	t_β	t_{ivx}	t_{ivx}^*	t_β	t_{ivx}	t_{ivx}^*	t_β	t_{ivx}	t_{ivx}^*
$\eta = 0$	0	29.5	4.6	7.9	10.7	5.4	5.2	5.4	5.3	5.2	28.7	5.3	5.7	11.1	5.3	5.3	5.3	4.7	4.7
	2	67.9	6.2	10.7	31.7	5.5	6	13.5	6.2	6	68.7	5.1	6	32.3	4.9	4.5	12.6	5.1	5
	5	97.7	8.3	63.9	73.5	9	14.6	46.8	9.4	9.5	97.9	5.3	23.4	74.5	5.2	6.2	46.4	5.4	5.4
	10	100	20	98.5	97.6	20.1	42.8	85.5	20.7	22.2	100	7	70.1	97.9	6.5	12.1	84.9	6.3	6.5
	15	100	33.4	100	99.8	33.2	70	96.9	32.4	37.1	100	8.9	93.3	99.9	9.1	24.4	97	9.2	9.3
	20	100	45.4	100	100	45.1	86.9	99.5	43.5	52.3	100	11.5	99.2	100	11.8	39.7	99.4	11.9	12.3
	30	100	65.7	100	100	64.7	98.6	100	63.6	77.9	100	18.9	100	100	19	68.6	100	18.7	20
$\eta = 0.2$	0	29.1	5.1	5.6	11.7	5	5.2	5.1	4.8	5.1	29.1	4.8	4.8	10.9	4.9	4.9	5.4	4.9	4.9
	2	68	6.8	8.4	32.7	7.1	7.7	12.8	8.1	8.1	69.1	5.3	5.7	31.9	5.5	5.4	12.8	5.7	5.5
	5	97.6	19.4	42	73.4	19.2	23.9	46.4	19.1	18.9	98.2	7.8	11	74.7	7.6	8.4	45.4	8.5	8.5
	10	100	47	87.8	97.4	45.8	59.3	85.6	43.3	42.7	100	14.9	31.8	98.1	15.6	20.7	85.7	15.8	15.5
	15	100	69.9	99.6	99.8	65.8	83.6	96.8	63.6	62.2	100	26	55.7	99.9	25.9	36.3	97.1	25.7	25.1
	20	100	84.7	100	100	81	95.3	99.5	78.7	77.6	100	37	76.1	100	36.5	53.5	99.5	35.5	34.8
	30	100	98.6	100	100	96.2	99.9	100	93.7	92.8	100	56.6	96.9	100	54.9	79.5	100	54.2	52.5
$\eta = 0.4$	0	29.4	5.5	5.5	10.6	4.7	5.1	4.9	5.1	5.2	28.9	5.1	5.2	11.1	5.1	5	4.9	4.6	4.7
	2	68.7	9.3	10.4	31.4	10.9	11	13.1	11.9	12	68.2	6.5	6.5	31.5	7.1	6.8	13.8	7.7	7.5
	5	97.9	39.9	50.7	73.7	36.9	39.7	46.3	35.6	35.2	98.1	16.6	19	73.6	16.8	17.4	45.7	17.2	17.1
	10	100	80.1	93.2	97.5	74	78.3	85.8	68.2	66.3	100	39.4	46.1	98	38.5	40.6	85	36.8	36.6
	15	100	97.8	100	99.9	91.8	94.7	96.9	87.1	84.7	100	61.5	71.7	99.9	59.8	63.9	97.2	56.3	55.6
	20	100	100	100	100	98.7	99.3	99.4	95.6	93.8	100	77.2	88.5	100	74.7	79.1	99.4	71.2	70.2
	30	100	100	100	100	100	100	100	99.7	99.3	100	96.4	99.8	100	93.4	96.5	100	91.1	89.7
$\eta = 0.6$	0	29.1	5.7	5.7	11.5	5.2	5.2	5.1	5.3	5.5	29.9	5.1	5.3	11.3	5.3	5.3	4.8	4.8	4.7
	2	68.5	12.6	13.5	31.6	16.8	16.9	12.8	17.8	17.8	69.1	9.2	9.3	31.5	10.5	10.5	12.6	11.4	11.4
	5	97.6	68.5	72.7	73.5	58.6	58.9	46	53.1	52.4	98.1	41.4	42.3	74.1	38.3	38.5	45.7	35.7	35.8
	10	100	99.6	99.7	97.7	92.1	92.3	85.6	84.9	84.4	100	82.8	84.2	97.7	75	75.3	85.5	70.3	70
	15	100	100	100	99.9	99.3	99.4	96.9	96.3	95.9	100	99	99.4	99.9	93.8	94.1	97.1	88.6	88.2
	20	100	100	100	100	100	100	99.4	99.1	98.9	100	100	100	100	99.2	99.3	99.4	96.2	95.9
	30	100	100	100	100	100	100	100	100	99.9	100	100	100	100	100	100	100	99.9	99.9
$\eta = 0.8$	0	28.9	5.7	5.5	11.8	5.6	5.5	5.7	5.3	5.4	29.3	5.4	5.2	11.2	4.8	4.9	5.3	5	5
	2	68.3	18.2	18.5	32.2	24.5	24.4	13.3	24.5	24.7	69.2	14.1	14	32.1	20.3	20.2	12.8	20.9	20.9
	5	97.7	92.3	92.5	73.1	73.8	73.8	46.7	64.9	64.5	98	84.1	84.4	73.1	67.8	67.7	46.6	59.7	59.7
	10	100	99.9	99.9	97.6	97.5	97.5	85.4	91.7	91.3	100	99.9	99.9	98.3	96.8	96.7	84.9	90	90
	15	100	100	100	99.9	99.8	99.7	97	98.3	98.2	100	100	100	99.9	99.9	99.9	96.5	97.7	97.7
	20	100	100	100	100	100	100	99.5	99.7	99.6	100	100	100	100	100	100	99.5	99.7	99.7
	30	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100

Note: t denotes the OLS-based test, t_{ivx} the IVX-based test and t_{ivx}^* the augmented IVX-based test of no predictability. The dependent variable is given by $y_t = \mu + \beta_T x_{t-1} + u_t$ with $\mu = 0$, the predictor is $x_{t-1} = \sum_{j=1}^{t-1} v_j$, the IVX instrument is generated with $\phi_T = 1 - 1/T^\eta$ and $z_0 = 0$, and ω is the correlation between u_t and v_t . The involved series are all demeaned, with the exception of the instrument z_t . OLS residuals are employed for the residual variance estimator required for computing standard errors. For further details see the text.

Moreover,

$$\frac{1}{T} \sum_{t=2}^T y_{t-1}^2 = \frac{T\beta_T^2}{T^2} \sum_{t=2}^T x_{t-2}^2 + \frac{2\beta_T}{T} \sum_{t=2}^T x_{t-2} u_{t-1} + \frac{1}{T} \sum_{t=2}^T u_{t-1}^2 \xrightarrow{p} \sigma_u^2$$

since $\beta_T = o(1)$ and $T\beta_T^2 = o(1)$ in all three cases,

$$\begin{aligned} \frac{1}{\sqrt{T}} \sum_{t=2}^T y_{t-1} u_t &= \frac{\beta_T}{\sqrt{T}} \sum_{t=2}^T x_{t-2} u_t + \frac{1}{\sqrt{T}} \sum_{t=2}^T u_{t-1} u_t \\ &= O_p \left(\max \left\{ \beta_T \sqrt{T}; 1 \right\} \right) = O_p(1), \end{aligned}$$

and

$$\begin{aligned} \frac{1}{T^2 \beta_T} \sum_{t=2}^T y_{t-1} x_{t-1} &= \frac{1}{T^2} \sum_{t=2}^T x_{t-2} x_{t-1} + \frac{1}{T^2 \beta_T} \sum_{t=2}^T u_{t-1} x_{t-1} \\ &= \frac{1}{T^2} \sum_{t=2}^T x_{t-2}^2 + o_p(1) \end{aligned}$$

since in all three cases $(\beta_T T)^{-1} \rightarrow 0$. Finally,

$$\begin{aligned} \sum_{t=2}^T y_{t-1} y_t &= \beta_T \sum_{t=2}^T y_{t-1} x_{t-1} + \sum_{t=2}^T y_{t-1} u_t \\ &= O_p \left(\max \left\{ \beta_T^2 T^2; \sqrt{T} \right\} \right). \end{aligned}$$

Standard IV algebra now indicates that

$$\hat{\beta}_{ivx}^* = \frac{\sum_{t=2}^T z_{t-1} y_t - \sum_{t=2}^T z_{t-1} y_{t-1} \left(\sum_{t=2}^T y_{t-1}^2 \right)^{-1} \sum_{t=2}^T y_{t-1} y_t}{\sum_{t=2}^T z_{t-1} x_{t-1} - \sum_{t=2}^T z_{t-1} y_{t-1} \left(\sum_{t=2}^T y_{t-1}^2 \right)^{-1} \sum_{t=2}^T y_{t-1} x_{t-1}};$$

also, it is easily shown that $\hat{\beta}_{ivx}^* - \beta_T = o_p(T^{-1/2})$ leading to consistency of the residual variance estimator $\hat{\sigma}_u^2 = T^{-1} \sum_{t=2}^T (y_t - \hat{\beta}_{ivx}^* x_{t-1})^2$. [Kostakis et al. \(2010\)](#) resort to OLS residuals, $\hat{\sigma}_u^2 = T^{-1} \sum_{t=2}^T (y_t - \hat{\beta}_T x_{t-1})^2$; they have e.g. the advantage of being consistent for $\eta = 0$ as well (for $\eta = 0$, the IVX residuals are not consistent). Then,

$$\begin{aligned} t_{ivx}^* &= \frac{\sum_{t=2}^T z_{t-1} u_t - \sum_{t=2}^T z_{t-1} y_{t-1} \left(\sum_{t=2}^T y_{t-1}^2 \right)^{-1} \sum_{t=2}^T y_{t-1} u_t}{\hat{\sigma}_u \sqrt{\sum_{t=2}^T z_{t-1}^2 - \sum_{t=2}^T z_{t-1} y_{t-1} \left(\sum_{t=2}^T y_{t-1}^2 \right)^{-1} \sum_{t=2}^T y_{t-1} z_{t-1}}} \\ &\quad + \beta_T \frac{\sum_{t=2}^T z_{t-1} x_{t-1} - \sum_{t=2}^T z_{t-1} y_{t-1} \left(\sum_{t=2}^T y_{t-1}^2 \right)^{-1} \sum_{t=2}^T y_{t-1} x_{t-1}}{\hat{\sigma}_u \sqrt{\sum_{t=2}^T z_{t-1}^2 - \sum_{t=2}^T z_{t-1} y_{t-1} \left(\sum_{t=2}^T y_{t-1}^2 \right)^{-1} \sum_{t=2}^T y_{t-1} z_{t-1}}} \end{aligned}$$

Table 2Rejection frequencies of no predictability tests under local alternatives $\beta_T = b/\tau$ when the regressor is near-integrated.

	b	$T = 100$									$T = 1000$								
		$\omega = -0.98$			$\omega = -0.5$			$\omega = 0$			$\omega = -0.98$			$\omega = -0.5$			$\omega = 0$		
		t_β	t_{lux}	t_{lux}^*	t_β	t_{lux}	t_{lux}^*	t_β	t_{lux}	t_{lux}^*	t_β	t_{lux}	t_{lux}^*	t_β	t_{lux}	t_{lux}^*	t_β	t_{lux}	t_{lux}^*
$\eta = 0$	0	7.8	5	6.1	6.2	5	4.7	5.7	5.3	5.3	8.9	5.3	5	5.9	4.8	4.9	5.1	4.9	5
	2	14.7	5	7.7	10	4.9	5	7.4	5.3	5.4	14.6	5	5.4	9.9	5.4	5.4	6.8	5.3	5.2
	5	33.5	5.9	8.4	24.5	5.7	5.2	18.5	5.3	5.4	32.6	5.2	4.9	23.8	4.8	4.9	16.8	5.3	5.3
	10	87.6	7.7	5.7	67.2	8.4	9.2	53.2	7.9	8.1	86.7	5.3	5.2	65.4	5.3	5.1	50.5	5.1	5.1
	15	99.9	12.2	39	94	12.4	19.3	82.7	13	14.5	100	5.8	8.3	93.5	5.5	6.2	81.5	6	6.1
	20	100	17.2	94	99.4	17.5	35.9	95.5	17.8	22	100	6.2	29.2	99.4	5.9	8	94.6	6	6
	30	100	34.4	100	100	34	79.6	99.9	34.5	48.7	100	7.7	92.5	100	8.2	17.5	99.8	7.7	7.8
$\eta = 0.2$	0	7.9	5.2	4.8	5.9	5.1	5	5.3	5.1	5.2	9.1	5.2	5.1	5.9	5	4.9	4.8	5.5	5.5
	2	14.4	5.4	4.9	10.3	6.1	5.6	7.3	5.7	5.9	14.8	4.9	4.6	10	5.1	5.3	6.8	5.2	5.2
	5	32.4	7.5	5.9	24.8	7.9	8.1	18.8	7.8	7.8	32.8	5.7	5.3	23.3	5.7	5.5	17.1	5	4.9
	10	87.1	15.2	15.4	67.4	15.9	17.2	52.2	16.3	15.8	86.6	7	6.6	65.2	6.9	7.2	51.4	6.8	6.7
	15	99.9	30	42.1	94	31	37.5	82.4	30.8	29.7	100	9.5	10.8	93.3	9	9.7	80.7	9.6	9.5
	20	100	50.6	82.1	99.4	48.6	63.1	95	46.5	45.4	100	12.5	17.8	99.4	13.4	15.7	94.9	12.8	12.5
	30	100	86.6	100	100	82.2	95.5	99.9	78.8	78.7	100	23.9	47.2	100	24.8	34.4	99.8	23.4	22.3
$\eta = 0.4$	0	8.6	5.3	5	6.2	5	5.3	5.6	5	5.2	8.9	5.1	4.9	5.8	4.8	4.8	4.8	4.8	4.9
	2	14.8	5.9	5.1	10.5	6.2	6.2	7.4	6.4	6.5	15.5	5.2	5.3	10.5	5.7	5.7	6.3	5.2	5.2
	5	32.6	10.3	8.8	25.6	11.1	11.2	19.3	11.6	11.7	32.5	6.9	6.5	24.2	7	6.8	17	7.3	7.3
	10	88.3	28.3	27.2	67.1	28.7	28.9	53.5	28.4	27.1	87	12.6	12.8	65.4	13.1	13.3	50.5	12.8	12.6
	15	100	59.1	63.1	93.9	55.6	58.6	81.8	52.6	50.2	100	23.3	24.5	93.8	23.2	24	80.7	23.1	22.6
	20	100	89.1	95.2	99.4	80.3	83.5	95.5	74.9	72	100	38.3	41.6	99.3	36.9	38.5	94.5	37.6	36.7
	30	100	99.7	100	100	97.9	98.9	99.9	95.5	93.7	100	73.9	83.3	100	70.3	74.4	99.8	67.3	65.6
$\eta = 0.6$	0	8.7	5.5	5.1	6.2	5.1	5.4	5.7	5.4	5.6	8.5	4.8	5	5.5	4.6	4.5	4.6	5.1	5.2
	2	14.6	6.3	6.3	10.1	6.4	6.5	7.6	7.1	7.2	14.9	5.8	5.6	9.3	6.1	6.2	7.2	6.1	6.2
	5	33.2	12.6	11.1	24	13.1	12.9	19.1	14.8	14.7	32.5	9.9	10.3	24.8	11.2	11.1	16.6	10.3	10.3
	10	86.8	39.8	37.5	66.7	40.4	40.3	53.3	40.5	39.3	87.3	26	25.5	65.7	26.8	26.6	50.9	27	26.8
	15	100	85.2	85.1	93.8	74.3	74.6	82.5	68	65.8	99.9	59.2	59.6	93.6	55.1	55.1	80.9	51	50.7
	20	100	98.8	99	99.4	92.2	92.4	95.4	86.5	84.5	100	89.5	90.5	99.5	80.1	80.6	94.8	73.6	73
	30	100	99.7	99.9	100	98.9	98.9	99.8	97.2	96	100	99.8	99.8	100	97.7	97.9	99.8	94.9	94.5
$\eta = 0.8$	0	8.7	5.4	5.4	6.4	5.9	5.9	5	5.2	5.4	8.6	5.2	5.1	6.3	5	4.9	4.9	4.8	4.8
	2	14.4	6.3	5.9	10.4	6.9	6.7	7	7	7.2	15.2	6.4	6.1	10.2	6.2	6.2	7.1	6.7	6.8
	5	33.2	13.7	12.5	24.9	15	15.1	18.5	16.2	16.2	33	12.5	12.5	23.8	13.2	13.2	17.1	13.8	13.7
	10	87.5	48.2	45.1	66.8	47	46.9	52.2	45.1	44.4	87	42.8	42.3	65.6	43	43	50.8	40.9	40.7
	15	100	92	91.2	94.1	79.4	79.1	82.5	72.1	70	100	90.2	89.9	93	76.1	76.2	80.2	69.1	69
	20	100	98.1	98	99.5	92.6	92.5	95.2	87	84.9	100	98.5	98.4	99.6	92.5	92.4	94.5	85.8	85.4
	30	100	99.2	99.2	100	97.5	97.2	99.8	95.4	93.7	100	99.5	99.5	100	98.2	98.1	99.8	96.4	96.2

Note: the predictor is $x_{t-1} = \sum_{j=1}^{t-1} (1 - c/\tau)^j v_{t-j}$ with $c = 10$; see Table 1 and the text for further details.

and it immediately follows that

$$\begin{aligned} \hat{\sigma}_u \sqrt{\frac{\sum_{t=2}^T Z_{t-1} u_t - \sum_{t=2}^T Z_{t-1} y_{t-1} \left(\sum_{t=2}^T y_{t-1}^2 \right)^{-1} \sum_{t=2}^T y_{t-1} u_t}{\sum_{t=2}^T Z_{t-1}^2 - \sum_{t=2}^T Z_{t-1} y_{t-1} \left(\sum_{t=2}^T y_{t-1}^2 \right)^{-1} \sum_{t=2}^T y_{t-1} Z_{t-1}}} \\ = \frac{\frac{1}{T^{1/2+\eta/2}} \sum_{t=2}^T Z_{t-1} u_t}{\sigma_u \sqrt{\frac{1}{T^{1+\eta}} \sum_{t=2}^T Z_{t-1}^2}} + o_p(1) \end{aligned}$$

$$t_{lux}^* = \frac{\sum_{t=2}^T Z_{t-1} u_t}{\sigma_u \sqrt{\sum_{t=2}^T Z_{t-1}^2}} + T^{1/2+\eta/2} \beta_T \frac{\frac{1}{T^{1+\eta}} \sum_{t=2}^T Z_{t-1} x_{t-1} - b \sigma_{uv} \frac{1}{\sigma_u^2} \frac{1}{T^2} \sum_{t=2}^T x_{t-2} x_{t-1}}{\sigma_u \sqrt{\frac{1}{T^{1+\eta}} \sum_{t=2}^T Z_{t-1}^2}} + o_p(1),$$

converging thanks to (4) and (5) to a standard normal distribution as required.

We then have along the same lines that

$$t_{lux}^* = \frac{\sum_{t=2}^T Z_{t-1} u_t}{\sigma_u \sqrt{\sum_{t=2}^T Z_{t-1}^2}} - \frac{T^2 \beta_T^2}{T^{1/2+\eta/2}} \frac{\sigma_{uv} \frac{1}{\sigma_u^2} \frac{1}{T^2} \sum_{t=2}^T x_{t-2} x_{t-1}}{\sigma_u \sqrt{\sum_{t=2}^T Z_{t-1}^2}} + o_p(1)$$

as required for the result, since $\frac{1}{T^2} \sum_{t=2}^T x_{t-2} x_{t-1} \Rightarrow \sigma_v^2 \int_0^1 B_c^2(s) ds$. For the second case we have correspondingly that

$$t_{lux}^* = \frac{\sum_{t=2}^T Z_{t-1} u_t}{\sigma_u \sqrt{\sum_{t=2}^T Z_{t-1}^2}} + T^{1/2+\eta/2} \beta_T \frac{\frac{1}{T^{1+\eta}} \sum_{t=2}^T Z_{t-1} x_{t-1}}{\sigma_u \sqrt{\frac{1}{T^{1+\eta}} \sum_{t=2}^T Z_{t-1}^2}} + o_p(1).$$

The case $\sigma_{uv} = 0$ follows analogously, thus concluding the derivations.

4. Finite-sample analysis

In order to assess the relevance of the asymptotic results in finite samples, series from (1) are replicated with $\beta_T = b/\tau$ with

$b \in \{0, 2, 5, 10, 15, 20, 30\}$. The number of Monte Carlo replications is 1000 for each studied case. The (size-distorted) OLS test is compared to the IVX test and the augmented IVX test, both computed with $a = 1$ and $\eta \in \{0, 0.2, 0.4, 0.6, 0.8\}$.

All three statistics are computed with demeaning; for both IVX statistics, only the dependent variable and the regressor are demeaned, but not the instrument. Both IVX-based tests are computed with residual variance estimator based on OLS residuals.

The experiment is conducted for two sample sizes, $T = 100$ and $T = 1000$; the correlation $\omega = \frac{\sigma_{uv}}{\sigma_u \sigma_v}$ causing distortions in the OLS-based statistic is set for both sample sizes to -0.98 , -0.5 and 0 . Although quite close to -1 , $\omega = -0.98$ is not an uncommon value in practice; see e.g. Breitung and Demetrescu (forthcoming, Section 5). The results are presented in Table 1 for $c = 0$, and in Table 2 for $c = 10$, to assess the effectiveness of the augmentation when the predictor is “closer” to stationarity.

For the random-walk predictor in Table 1, we expectedly observe that the OLS-based test is only keeping size when $\omega = 0$. At the same time, size control is good for both the IVX and the augmented IVX tests; the behavior in terms of size is essentially the same for IVX and augmented IVX over all parameter constellations. (For larger η , this is a consequence of not demeaning z_t . Unreported simulations, available upon request, show that demeaning z_t does lead to distortions: e.g. already for $\eta = 0.6$, rejection frequencies above 10% are common for large ω and small T .)

In terms of power, the rejection frequencies for OLS under the alternative approach relatively fast 100%; this is partly due to the oversizedness, but is also observed for $\omega = 0$, illustrating that the OLS-based test has power in $1/T$ neighborhoods of the null for near-integrated regressors; there are minor differences between $T = 100$ and $T = 1000$, due to Monte Carlo variability. For the two IVX procedures, the power diminishes when T increases, capturing the fact that local power is not given for IVX-based procedures in such $1/T$ neighborhoods. For $\omega = 0$, the power of the two is practically the same, as predicted by the asymptotics. When $\omega \neq 0$, the differences between the two IVX variants are quite spectacular for $\eta = 0$ and $\eta = 0.2$; even for $\eta = 0.4$ one can still see an advantage for the augmented IVX. Moreover, the power reduction when T increases is minimal for the augmented IVX test, but serious for the IVX test—explained by the higher rate $T^{-\frac{3-\eta}{4}}$ of the augmented IVX procedure. Also, the larger ω in the absolute value, the greater the advantage of augmenting the IVX regression. For $\eta = 0.6$ or $\eta = 0.8$, however, the two IVX variants are throughout practically indistinguishable, equivalence predicted by the asymptotic result.

In the case of a near-integrated regressor with $c = 10$, the findings are qualitatively the same. In particular, for $\eta = 0, 0.2, 0.4$ and $\omega \neq 0$, augmented IVX has higher power. The magnitude of the differences to IVX is however smaller than in the random walk case, and especially visible for $b = 15, 20, 30$. This is not surprising with lower overall power due to the reduced strength of the signal (which also implies overall size distortions of the OLS-based test).

To sum up, the boost in local power is given as long as there is endogeneity (in the predictive regression sense) and $\eta < 1/3$. For $\eta > 1/3$, the limiting behavior of the modified t statistic is the same as for the initial IVX regression so nothing is lost by using the augmented IVX version.

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